

QUESTION 1

What is the & used for in C++?

- A. To get the address of a variable B. To indicate a reference parameter to a function C. To follow a pointer
D. Both A and B E. None of these

QUESTION 2

If x has the value 5, what is the value of the expression x++?

- A. 0 B. 4 C. 5 D. 6 E. None of these

QUESTION 3

Which of the following is *not* used to store floating point data?

- A. float B. double C. long D. long double E. None of these

QUESTION 4

Which of the following make an `apstring` which holds the string "hello"?

- I. `apstring a = "hello";` II. `apstring a="he"; a+="llo";` III. `apstring a("hello");`

- A. I only B. II only C. III only D. II and III E. I, II, and III

QUESTION 5

Which of the following is true about methods in a `private` section of a class declaration?

- A. They can be invoked inside other methods in a `private` section of the same class
B. They can be invoked inside other methods in a public section of the same class
C. They can not be invoked in the `main()` function
D. They can not be invoked inside functions declared outside the class
E. All of the above

QUESTION 6

What is returned by the function call `mystery("672",5)`?

- A. 20 B. 21 C. 22
D. 23 E. None of these

```
int mystery(apstring s, int x) {
    if (s.length()==0) return x;
    else
        return
            mystery(s.substr
                    (1,s.length()-1),
                    x+s[0] - '0');
}
```

For the following questions, give the output of the code to the right for the given input. In the output, * represents any character which is not alphanumeric.

QUESTION 7

abcde

- A. abcde
- B. aabbccdde
- C. eeeee
- D. fffff
- E. None of these

QUESTION 8

APPLE
PIE

- A. HVUPHRJE
- B. IWVQI*RJE
- C. JXWRJ**RJE
- D. KYVSK***RJE
- E. None of these

QUESTION 9

0 0
1 3 5
24

- A. *****9*9*9**34
- B. 9*77*7*734
- C. 7666734
- D. *****
- E. None of these

QUESTION 10

$CAB_{16} - ACE_{16} =$

- A. $2AC_{16}$
- B. AAA_{16}
- C. $1DD_{16}$
- D. 914_{16}
- E. None of these

```
int main() {
    apstring s;
    char ch;
    while (cin>>ch) {
        s+=ch;
        for(int i=0;i<s.length()-1;++i)
            s[i]++;
    }
    cout << s;
}
```

QUESTION 11

Which of these correctly fills **<*1>** in the code to the right to check that `apmatrix a` and `apmatrix b` have the same number of rows and the same number of columns?

- A. `a.numrows=b.numrows=a.numcols=b.numcols`
- B. `a.numrows==a.numcols || b.numrows=b.numcols`
- C. `a.numrows()==b.numrows() && a.numcols()==b.numcols()`
- D. `a.numrows()+a.numcols()==b.numrows()+b.numcols()`
- E. None of these

```
template <class itemType>
apmatrix<itemType>
    add(apmatrix<itemType> a,
        apmatrix<itemType> b) {
    apmatrix<itemType> c;
    if (<*1>) {
        <*2>
        for(int i=0;i<c.numrows();++i)
            for(int j=0;j<c.numcols();++j)
                c[i][j]=a[i][j]+b[i][j];
    }
    return c;
}
```

QUESTION 12

Which member function of `apmatrix` is used in **<*2>** to change the size of `c` to have the number of rows and columns of `a`?

- A. `operator=` B. `operator[]` C. `resize()`
- D. copy constructor E. None of these

QUESTION 13

Assume that **<*1>** and **<*2>** are filled in correctly. What matrix is returned by `add(A, B)` where `A` and `B` are initialized as below?

A:	1	2	B:	3	4
	3	4		7	8

A:	<table border="1"><tr><td>4</td><td>6</td></tr><tr><td>10</td><td>12</td></tr></table>	4	6	10	12
4	6				
10	12				

B.	1	2	3	4
	3	4	7	8

C.	<table><tr><td>32</td></tr></table>	32
32		

D. an empty matrix

E. None of these

QUESTION 14

What is returned by chop("banana pudding", 'a')?

- A. "anan" B. "banana pudding"
- C. "anana" D. empty string
- E. None of these

QUESTION 15

What is returned by chop("banana pudding", 'z')?

- A. "anan" B. "banana pudding"
- C. "anana" D. empty string
- E. None of these

QUESTION 16

Which of the following sets the first entry in the array of favorites for cookbook c to 237?

- A. c.favorites==237
- B. c.favorites[0]=237
- C. cookbook.favorites[0]==237
- D. c[3][0]=237
- E. None of these

```
apstring chop(apstring & s,
              char ch) {
    int beg,end;
    beg=end=s.find(ch);
    if (beg>=0 && beg<s.length())
        for (int i=beg;i<s.length();
              ++i)
            if (s[i]==ch) end=i;
    return s.substr(beg,end-beg);
}
```

QUESTION 17

Suppose the size of int is four times the size of char. What is sizeof(mystruct), assuming no padding is needed?

- A. 13 B. 52
- C. 26 D. 4
- E. None of these

```
enum style {AMERICAN, CHINESE,
            EHTIOPAN, INDIAN, ITALIAN,
            MEXICAN};
```

```
struct cookbook {
    int year;
    int numpages;
    style s;
    int favorites[10];
}
```

QUESTION 18

Which of the following will output the minimum of x and y?

- A. cout << (x<y)?x:y;
- B. cout << if (x<y) x; else y;
- C. switch(x<y) {
 case true: cout << x;
 case false: cout << y;
 }
- D. Both A and C
- E. A, B, and C

QUESTION 19

What should replace **<*1>** to include the definition of the fraction class?

- A. `<fraction>`
- B. `<fraction.h>`
- C. `"fraction"`
- D. `"fraction.h"`
- E. None of these

QUESTION 20

The private method `reduce()` is supposed to reduce a fraction to lowest terms. What should fill **<*2>** as an argument to `reduce()`?

- A. a fraction
- B. an integer
- C. nothing
- D. methods can't be private
- E. None of these

For the remaining questions, assume **<*1>** and **<*2>** are filled in correctly.

QUESTION 21

Which of these declares a `fraction` representing two thirds?

- A. `fraction f(2/3);`
- B. `fraction f(2./3.);`
- C. `fraction f(3/2);`
- D. `fraction f(3./2.);`
- E. None of these

QUESTION 22

Suppose `f1` represents the fraction two thirds and `f2` represents the fraction one fourth. What are the values of the numerator and the denominator of the fraction returned by `f1.sub(f2)`?

- A. `n=1, d=6`
- B. `n=8, d=3`
- C. `n=9, d=12`
- D. `n=7, d=12`
- E. None of these

```
fraction.h:
class fraction {
public:
    fraction(int num,int den);
    fraction add(fraction b);
    fraction sub(fraction b);
    fraction mul(fraction b);
    fraction div(fraction b);
private:
    int n, d;
    void reduce(<*2>);
};

fraction.cpp:
#include <*1>

int lcm(int x,int y) {
    // returns least common multiple
}
int gcd(int x, int y) {
    // returns greatest common
    // divisor
}

fraction::fraction(int num,
                    int den){
    n=num; d=den;
    reduce();
}

fraction fraction::add(fraction
                       b){
    int l=lcm(d,b.d);
    fraction c(l/d*n+l/b.d*b.n,l);
    return c;
}

fraction fraction::sub(fraction
                       b){
    fraction c(-b.n,b.d);
    return add(c);
}

fraction fraction::mul(fraction
                       b){
    fraction c(n*b.n,d*b.d);
    c.reduce(); return c;
}

fraction fraction::div(fraction
                       b){
    fraction c(b.d,b.n);
    return mul(c);
}

void fraction::reduce(<*2>) {
    g=gcd(n,d);
    n/=g; d/=g;
}
```

QUESTION 23

What could replace `x%2==0` in the code to the right without changing the behavior of the code for positive input?

- A. `x%2-1`
- B. `x/2==0`
- C. `x/2!=0`
- D. `x`
- E. None of these

```
apstack<int> s;
apqueue<int> q;
int x;

while (cin >> x) {
    if (x%2==0) s.push(x);
    else q.enqueue(x);
}
```

```
while (!s.isEmpty()) {
    cout << s.top() << ' ';
    s.pop();
}
```

```
while (!q.isEmpty()) {
    cout << q.front() << ' ';
    q.dequeue();
}
```

QUESTION 24

What is output by the code to the right on this input?

1 2 3 4 9 7 5 6 8

- A. 1 2 3 4 9 7 5 6 8
- B. 8 6 5 7 9 4 3 2 1
- C. 8 6 4 2 1 3 9 7 5
- D. 5 7 9 3 1 2 4 6 8
- E. None of these

QUESTION 25

Which of the following expressions is true?

- A. `one < two < ten`
- B. `ten < two < one`
- C. `one < ten < two`
- D. `two < ten < one`
- E. None of the above

```
apstring one("1");
apstring two("2");
apstring ten("10");
```

QUESTION 26

One version of the assignment operator definition for the `apstring` class has an argument `const apstring & str`. What does this mean about a call to the assignment operator?

- A. The string on the left side is passed by reference, but cannot be modified
- B. The string on the right side is passed by reference, but cannot be modified
- C. The string on the left side is passed by value, so the original value doesn't change
- D. The string on the right side is passed by value, so the original value doesn't change
- E. None of these.

QUESTION 27

In the code to the right, what is output by the first call to `f()`?

- A. 84 B. 83 C. 74
D. 73 E. None of these

```
int f(int x) {
    static int y=3;
    x+=y++;
    cout << x << y;
}
```

QUESTION 28

In the code to the right, what is output by the second call to `f()`?

- A. 84 B. 83 C. 74
D. 73 E. None of these

```
int main() {
    int z=4;
    f(z);
    f(z);
}
```

QUESTION 29

Which library defines a constant for the largest integer?

- A. `ctype.h` B. `max.h` C. `limits.h` D. `int.h` E. None of these

QUESTION 30

Which of these expressions is represented by the truth table below?

a	b	c
0	0	1
0	1	0
1	0	0
1	1	0

- A. `!a || !b` B. `!(a&&b)` C. `a || b`
D. `!a&&!b` E. None of these

QUESTION 31

Which of these expressions sets `y` to point to `x[3]`?

- A. `y = x+3;`
B. `*y = *x[3];`
C. `y = &x+3;`
D. `*y = x[3];`
E. None of these

```
int x[10];
int * y;
```

QUESTION 32

What advantage does a sequential search algorithm have over a binary search?

- A. It runs faster on sorted lists
- B. It runs faster when the item is not in the list
- C. It sorts the list as it goes
- D. It works on unsorted lists
- E. None of these

QUESTION 33

Which of the following data structures for holding a list of items allows an insertion to be made in the middle in $O(1)$ time, assuming that there is already a pointer to the place where the insertion is to occur?

- A. linked list
- B. apvector
- C. C style array
- D. More than one of these
- E. None of these

QUESTION 34

Which of the following is a possible output of main on the input below?
1 3 5 7 9 11

- A. ABcdEF
- B. vDFkmE
- C. BWfsJO
- D. AZazAZ
- E. More than one of these

```
char up(int x, bool b) {
    if (b)
        return (x%26+'A');
    else
        return ('Z'-x%26);
}

char down(int x, bool b) {
    if (b)
        return (x%26+'a');
    else
        return ('z'-x%26);
}
```

QUESTION 35

Which of the following is a possible output of main on the input below?
2 4 6 8

- A. CVgr
- B. XetI
- C. cvGR
- D. xETi
- E. More than one of these

```
int main() {
    int y;
    int cycle;
    while (cin >> y) {
        switch((cycle++)%4) {
            case 0: cout<<up(y,true);
                    break;
            case 1: cout<<up(y,false);
                    break;
            case 2: cout<<down(y,true);
                    break;
            case 3: cout<<down(y,false);
                    break;
        }
    }
}
```

QUESTION 36

Starting with the vector **v** below, what will **v** look like after the inner loop has run twice?

1	8	3	2	4	7	9
---	---	---	---	---	---	---

A.

1	7	3	2	4	8	9
---	---	---	---	---	---	---

B.

1	3	2	4	7	8	9
---	---	---	---	---	---	---

C.

1	2	3	8	4	7	9
---	---	---	---	---	---	---

D.

1	2	3	4	7	8	9
---	---	---	---	---	---	---

E. None of these

```
void sort(apvector<int> & v) {
    int max;
    for (int i=v.length();i>=1;--i){
        max=i-1;
        for (int j=i-2; j>=0; --j)
            if(v[j]>v[max]) max=j;
        swap(v[i-1],v[max]);
    }
}
```

QUESTION 37

What sort is implemented by the code to the right?

A. MergeSort B. QuickSort C. Selection

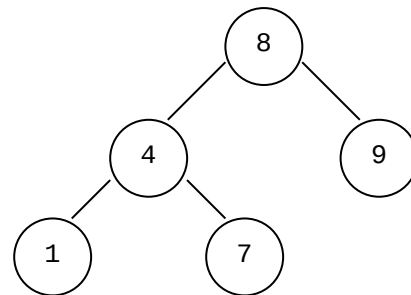
D. Insertion E. None of these

QUESTION 38

Which of the following is a preorder traversal of the tree to the right?

A. 1 4 7 8 9 B. 8 4 9 1 7 C. 1 7 4 9 8

D. 9 8 7 4 1 E. None of these

**QUESTION 39**

If **n** is 5, what is the value of **COUNT** after running the code to the right?

A. 36 B. 25 C. 10

D. 5 E. None of these

```
int count = 0
for (int i=0; i<n; ++i)
    for (int j=0; j<n; ++j)
        count++;
```

QUESTION 40

What is the running time of the code to the right? Choose the smallest correct answer.

A. $O(n^2)$ B. $O(n \log n)$ C. $O(n)$

D. $O(\log n)$ E. None of these

AP QUICK REFERENCE

NOTE: The front and back of this page includes reference information for the AP classes. This is the same reference information that students use during the AP exam. You may use this reference sheet during the contest. You may detach this page from the test booklet.

Quick Reference for apstring

extern const int npos; // used to indicate not a position in the string

class apstring

// public member functions

// constructors/destructor

```
apstring(); // construct empty string ""
apstring(const char * s); // construct from string literal
apstring(const apstring & str); // copy constructor
~apstring(); // destructor
```

// assignment

```
const apstring & operator= (const apstring & str); // assign str
const apstring & operator= (const char * s); // assign s
const apstring & operator= (char ch); // assign ch
```

// accessors

```
int length() const; // number of chars
int find(const apstring & str) const; // index of first occurrence of str
int find(char ch) const; // index of first occurrence of ch
apstring substr(int pos, int len) const; // substring of len chars, starting at pos
const char * c_str() const; // explicit conversion to char *
```

// indexing

```
char operator[ ](int k) const; // range-checked indexing
char & operator[ ](int k); // range-checked indexing
```

// modifiers

```
const apstring & operator+= (const apstring & str); // append str
const apstring & operator+= (char ch); // append char
```

// The following free (non-member) functions operate on strings

// I/O functions

```
ostream & operator<< ( ostream & os, const apstring & str );
istream & operator>> ( istream & is, apstring & str );
istream & getline( istream & is, apstring & str );
```

// comparison operators

```
bool operator== ( const apstring & lhs, const apstring & rhs );
bool operator!= ( const apstring & lhs, const apstring & rhs );
bool operator< ( const apstring & lhs, const apstring & rhs );
bool operator<= ( const apstring & lhs, const apstring & rhs );
bool operator> ( const apstring & lhs, const apstring & rhs );
bool operator>= ( const apstring & lhs, const apstring & rhs );
```

// concatenation operator +

```
apstring operator+ ( const apstring & lhs, const apstring & rhs );
apstring operator+ ( char ch, const apstring & str );
apstring operator+ ( const apstring & str, char ch );
```

AP QUICK REFERENCE CONT.

Quick Reference for apvector and apmatrix

```
template <class itemType>
class apvector

// public member functions

// constructors/destructor
apvector(); // default constructor (size==0)
apvector(int size); // initial size of vector is size
apvector(int size, const itemType & fillValue); // all entries == fillValue
apvector(const apvector & vec); // copy constructor
~apvector(); // destructor

// assignment
const apvector & operator= (const apvector & vec);

// accessors
int length() const; // capacity of vector

// indexing
itemType & operator[ ](int index); // indexing with range checking
const itemType & operator[ ](int index) const; // indexing with range checking

// modifiers
void resize(int newSize); // change size dynamically; can result in losing values
```

```
template <class itemType>
class apmatrix

// public member functions

// constructors/destructor
apmatrix(); // default size is 0 x 0
apmatrix(int rows, int cols); // size is rows x cols
apmatrix(int rows, int cols, const itemType & fillValue); // all entries==fillValue
apmatrix(const apmatrix & mat); // copy constructor
~apmatrix( ); // destructor

// assignment
const apmatrix & operator = (const apmatrix & rhs);

// accessors
int numrows() const; // number of rows
int numcols() const; // number of columns

// indexing
const apvector<itemType> & operator[ ](int k) const; // range-checked indexing
apvector<itemType> & operator[ ](int k); // range-checked indexing

// modifiers
void resize(int newRows, int newCols); // resizes matrix to newRows x newCols
// (can result in losing values)
```

Computer Science Answer Key

UIL Invitational B 2003

1. D	11. C	21. E	31. A
2. C	12. C	22. D	32. D
3. A	13. A	23. A	33. A
4. E	14. A	24. C	34. C
5. E	15. D	25. C	35. E
6. A	16. B	26. B	36. A
7. C	17. B	27. C	37. C
8. A	18. B	28. A	38. E
9. C	19. D	29. C	39. B
10. C	20. C	30. D	40. A